

Problem Set 8

Submission:

Friday, 24/04/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.

Only problems marked with (*) will be graded.

1. Joint distribution and independence

- (a) Let $A \subseteq \mathbb{R}^2$ be the closed triangle spanned by the points $(0, 0)$, $(0, 1)$ and $(1, 0)$. We say that the random vector $(X, Y) : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$ follows a continuous uniform distribution on A , if the joint law of X, Y , denoted by $\mathbf{P}_{(X,Y)}$, has a multivariate probability density function

$$f_{X,Y}(x, y) = c \cdot \mathbb{1}_A(x, y) = \begin{cases} c, & (x, y) \in A, \\ 0, & (x, y) \notin A. \end{cases}$$

We write $(X, Y) \sim \mathcal{U}(A)$.

- Calculate the value of c . Explain briefly why the law of (X, Y) is called “uniform distribution”.
 - Compute the marginal probability densities f_X and f_Y .
 - Are X and Y independent? Give a precise mathematical argument.
 - Calculate the covariance $\text{Cov}[X, Y]$.
- (b) Let $X, Y \sim \mathcal{N}(0, 1)$ be independent random variables. We define the random variable Z by setting

$$Z = \text{sign}(Y) \cdot X = \begin{cases} X, & \text{if } Y > 0 \\ -X, & \text{if } Y \leq 0. \end{cases}$$

- Determine the law of Z .
- Calculate the probability $\mathbf{P}[X + Z = 0]$. Is the random variable $W = X + Z$ discrete, continuous or neither?
- Are X and Z independent? Give a precise mathematical explanation.

2. (*) Bertrand's paradox

[4 Points]

We consider an equilateral triangle inscribed in a circle of radius 1. Suppose a chord of the circle is chosen at random. What is the probability that the chord is longer than a side of the triangle? For solving this try the following probability models:

- (a) The “random endpoints” method: Choose two uniform random points on the circumference of the circle and draw the chord joining them, i.e., let $U, V \sim \mathcal{U}((0, 1))$, define

$$X = e^{U2\pi i} = (\cos(2\pi U), \sin(2\pi U)), \quad Y = e^{V2\pi i} = (\cos(2\pi V), \sin(2\pi V))$$

and take the chord connecting X and Y .

- (b) The “random radius” method: Choose a radius of the circle, choose a uniform point on the radius and construct the chord through this point and perpendicular to the radius, i.e., choose a radius and choose $R \sim \mathcal{U}((0, 1))$, take the point on the radius that is at distance R from the center and the chord will be the only chord perpendicular to this point.

- (c) The “random midpoint” method: Choose a point uniformly anywhere within the circle (see Problem 1.(a), but with A replaced by a disk) and construct a chord with the chosen point as its midpoint, i.e., and take the chord whose midpoint is (X, Y) .
- (d) Is this a contradiction?

3. Convolution

- (a) On normal distributions:

- i. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ independent random variables with $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ with $\mu_1, \mu_2 \in \mathbb{R}$ and $\sigma_1, \sigma_2 > 0$. Show that

$$X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Hint: First explain why it suffices to show this for $\mu_1 = \mu_2 = 0$ and $\sigma_1 = 1$. Then complete the square in the exponent.

- ii. Now let $n \in \mathbb{N}$ and let $X_1, \dots, X_n : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ i.i.d. random variables with $X_1 \sim \mathcal{N}(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ and $\sigma > 0$. Consider the mean value $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Determine the law of the random variable

$$Z_n = \sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}.$$

- (b) On Poisson distributions:

- i. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ independent random variables with $X \sim \text{Pois}(\lambda)$ and $Y \sim \text{Pois}(\mu)$, where $\lambda, \mu > 0$. Show that

$$X + Y \sim \text{Pois}(\lambda + \mu).$$

- ii. Give a concrete counterexample showing that $X + Y$ does in general *not* follow a Poisson distribution, if we drop the assumption of independence in part i. of the problem.

4. (*) The student- t -distribution

[4 Points]

Suppose that Z has the standard normal distribution, V has the χ^2 distribution with n degrees of freedom, and that Z and V are independent. The χ^2 -distribution with n degrees of freedom is defined as the distribution of $\sum_{i=1}^n X_i^2$ where $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. Let

$$T = \frac{Z}{\sqrt{V/n}}.$$

You will show that T has p.d.f. given by

$$f(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2} \quad t \in \mathbb{R},$$

where we define $\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx$, for $t > 0$.

- (a) Show that V has the probability density function $f_V(x) = c_n x^{n/2-1} e^{-x/2} \mathbb{1}_{\{x \geq 0\}}$, with an appropriate $c_n \in \mathbb{R}$.

Hint: Calculate c_n first, then use induction.

- (b) Find the joint probability density function of (T, V) .

Hint: As always, look at the (joint) cumulative distribution function $F_{T,V}(t, v)$.

- (c) Find the probability density function of T .